Opioids Project - Data Science Memo

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GitHub Repository

0.1 Stakeholders

This memo is written for Policy Analysts and Officials at the United States Drug Enforcement Agency and the United States Department of Health and Human Services (HHS). The intended reader has strong policy expertise and moderate technical familiarity with public health metrics, but may not necessarily be a data scientist.

0.2 Executive Summary

The United States has experienced a rapid rise in prescription opioid use and overdose deaths since the early 2000s, creating a serious public health crisis and a pressing issue for policymakers. This memo uses Florida as a case study to understand whether prescription-regulating policies can reduce opioid misuse and overdose deaths. Florida became an early hotspot for prescription opioid misuse and responded by implementing a series of prescribing regulations beginning in 2010. By evaluating the effects of these policies in Florida, this memo provides national policymakers with evidence about which strategies help manage opioid use. This analysis helps address the broader problem by identifying which types of interventions meaningfully reduce overdose deaths and opioid misuse.

To evaluate the impact of Florida's policies, we analyze county-level data from 2006 to 2015 using a difference-in-difference design that compares Florida's trends to those of similar states that did not implement comparable regulations. This approach helps separate the effect of Florida's policy from broader national forces that were influencing opioid supply and overdose mortality during the same period. Florida is compared to Ohio, Alabama, Georgia, and Kentucky for prescription opioid shipments, and to Colorado, Michigan, and Pennsylvania for overdose deaths.

Our findings show that Florida's 2010 regulations produced a large and immediate reduction in prescription opioid shipments. Overdose deaths also declined in the first few years after the policy while deaths in comparison states continued to rise. Although Florida's overdose mortality eventually increased again as illicit fentanyl became more widespread, Florida remained below the control states for most of the post-policy period, indicating that the regulations had a meaningful effect even as national conditions shifted.

Florida's experience shows that prescribing regulations can reduce opioid availability but need to be combined with other efforts to lower overdose deaths. HHS can use this evidence to guide national

strategies focused on treatment and prevention, while the DEA can use it to strengthen oversight of the drug supply chain. Together, these agencies can apply lessons from Florida to improve the national response to the opioid crisis.

0.3 Decisions To Be Made

Based on this analysis, the US Department of Health and Human Services and the Drug Enforcement Administration must decide how to apply lessons from Florida's past policies to the current opioid epidemic. The findings suggest that prescription-regulating policies can meaningfully reduce the supply of opioids and may help slow the growth of overdoses, so both agencies should consider supporting similar policies where appropriate.

At the same time, both agencies must strengthen national monitoring and response to illicit opioids such as fentanyl, which appear to drive overdose increases even when prescription supply falls. Improving surveillance, enforcement, and harm-reduction strategies will be essential for addressing this part of the crisis.

0.4 Background and Research Questions

Over the past two decades, the United States has faced a severe opioid crisis marked by widespread prescription opioid misuse, rising addiction, and increasing overdose deaths. Florida became one of the earliest and most extreme hotspots. By 2010, 98 of the 100 highest-volume oxycodone-dispensing physicians in the country practiced in Florida, and hundreds of loosely regulated pain clinics were distributing large quantities of opioids with little medical oversight.

For Florida officials and federal agencies, this created an urgent policy problem. Excessive and inappropriate prescribing was driving addiction and preventable deaths, and state regulators had the authority to intervene in how pain clinics operated, how physicians dispensed controlled substances, and how opioid shipments were monitored.

Beginning in 2010, Florida introduced a series of major prescribing regulations. These included mandatory pain-clinic registration, enforcement operations targeting pill mills, limits on physician dispensing, creation of a statewide Prescription Drug Monitoring Program, required reporting from dispensers, and tighter oversight of drug distributors. These policies were intended to reduce inappropriate prescribing, though some worried that restricting prescription opioids might push individuals toward more dangerous illicit opioids such as heroin and fentanyl.

Using Florida as a case study, this memo examines a broader policy question: How effective are opioid prescribing regulations at controlling opioid use in the US? Specifically, we evaluate the impact of Florida's 2010 regulations on 1) Prescription opioid shipments (measured in morphine milligram equivalents), and 2) Opioid overdose deaths.

By analyzing how these outcomes changed in Florida relative to similar states, this memo provides evidence that can guide future state and national strategies for controlling opioid use. We hypothesize that the regulations substantially reduced prescription opioid availability but was not as effective at reducing the overall rate of overdose deaths in Florida.

0.5 Methodology

The goal of this analysis is to determine whether Florida's 2010 regulations actually reduced the availability of prescription opioids and overdose deaths. To answer this, we use a difference-in-difference approach that compares changes in outcomes in Florida to changes in similar states that did not implement comparable regulations at the same time. This approach helps isolate the impact of the policy from unrelated factors that affected all states simultaneously but are not under the state's control.

Our analysis uses county-level data from 2006 to 2015 on two outcomes:

1. Prescription opioid shipments (measured in morphine-milligram equivalents) per 1,000 people, which standardize different opioids into a common potency scale.

2. Opioid overdose deaths per 100,000 people.

The data was gathered from federal datasets: prescription shipments from the DEA's ARCOS system, overdose mortality from US Vital Statistics, and annual county populations from the US Census. Since overdose data are censored in small counties, we limit the analysis for overdose deaths to counties with populations above 350,000.

A simple pre-and-post policy comparison within Florida would not provide reliable evidence, because many factors outside of state policy were changing during this period. National medical guidelines, federal enforcement efforts, economic shifts, and the growing presence of illicit fentanyl all influenced opioid supply and overdose trends across the country. Any of these forces could cause Florida's outcomes to rise or fall even if the state had made no policy changes. To address this concern, we use a difference-in-difference approach. This method compares Florida's changes before and after the policy to the changes in similar states. By removing the background changes that were happening everywhere, this approach isolates the portion of Florida's post-policy shift that is more likely to reflect the impact of the regulations. Although no observational method can remove all possible confounding factors, a difference-in-difference design provides a stronger basis for causal interpretation than a simple pre-post comparison.

A key requirement of difference-in-difference analysis is that Florida must be compared to states with similar pre-policy trends. To satisfy this, we identify sets of control states for both outcomes whose 2006-2009 trends closely match Florida's. This ensures that any post-2010 differences in the outcomes between Florida and the controls is due to Florida's policies rather than pre-existing differences. Prior to the policy change, both Florida and the control states exhibit similar upward trends in both measures, indicating that the groups are comparable and supporting the use of a difference-in-difference framework. The control states selected for prescription shipments are Ohio, Alabama, Georgia, and Kentucky; those for overdose deaths are Colorado, Michigan, and Pennsylvania. We then compare Florida's changes in opioids shipments and overdose deaths to those of the selected control states before and after the 2010 regulations. See Appendix for a complete explanation of control state criteria.

By examining both prescription opioid availability and overdose mortality, this approach addresses the stakeholder's core question: how well do prescribing regulations help control opioid use and reduce overdose deaths. If Florida's outcomes decline more sharply than those in comparable states after 2010, this suggests that the regulations were effective in reducing over prescribing. If overdose deaths do not fall as much as prescription shipments, the results indicate that additional policies may be needed to reduce deaths more effectively.

0.6 Results

We first analyze the effectiveness of the policies in reducing the amount of prescription opioid shipments, measured through the MME (morphine milligram equivalents) per 1,000 people.

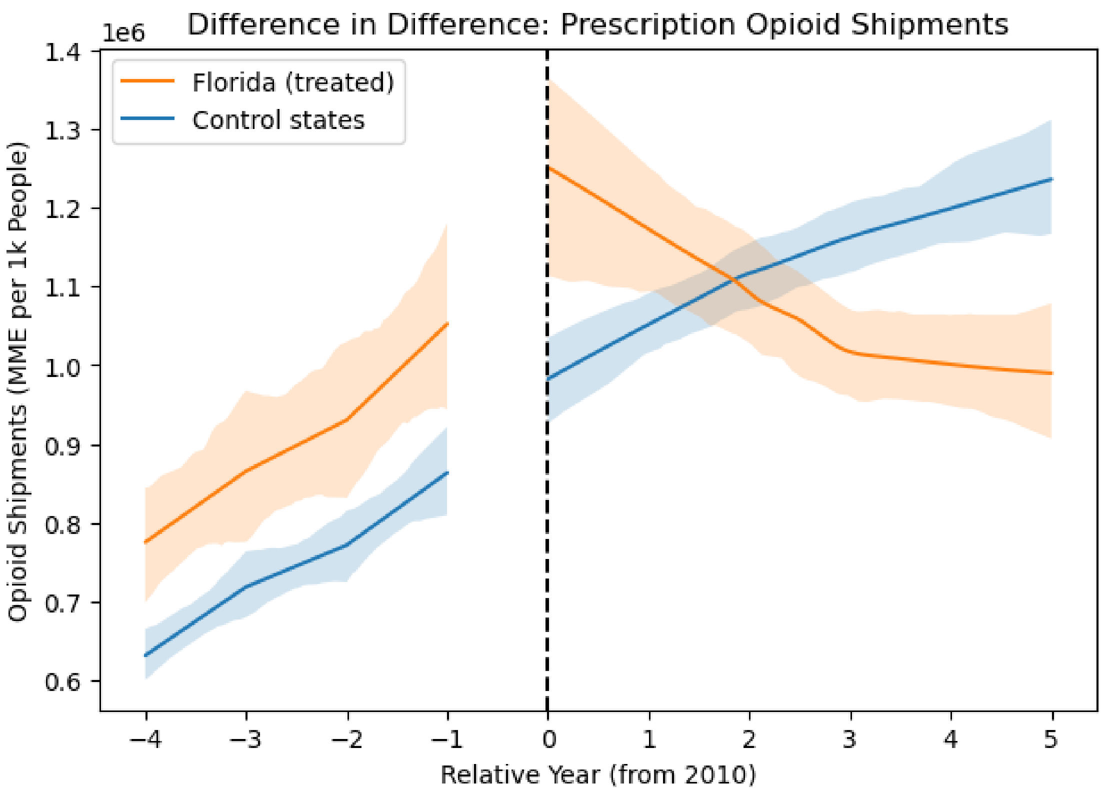

Figure 1. Difference in Difference: Opioid Shipments

Figure 1 shows how the prescription opioid shipments per 1,000 people changed in Florida compared to the selected control states- Ohio, Alabama, Georgia, and Kentucky- before and after Florida's 2010 prescribing regulations.

Immediately after the regulations were implemented (relative year 0), Florida's opioid shipments decline sharply and continue decreasing, dropping far more steeply than in control states. This pattern begins in the first post-policy year and continues for several years, whereas the control states experience only a modest reduction over the next five years. Florida experienced a larger reduction in shipments than the control states, as shown by the narrowing gap between Florida and the controls. This indicates a large reduction in the supply of prescription opioids within Florida due to the policies.

The results strongly suggest that Florida's 2010 opioid regulations were effective at reducing the supply of prescription opioids entering the state, achieving the intended goal of reducing inappropriate prescribing and opioid use.

Next, we examine the effectiveness of the policies in reducing the number of overdose deaths, measured through the number of overdose deaths per 100,000 people.

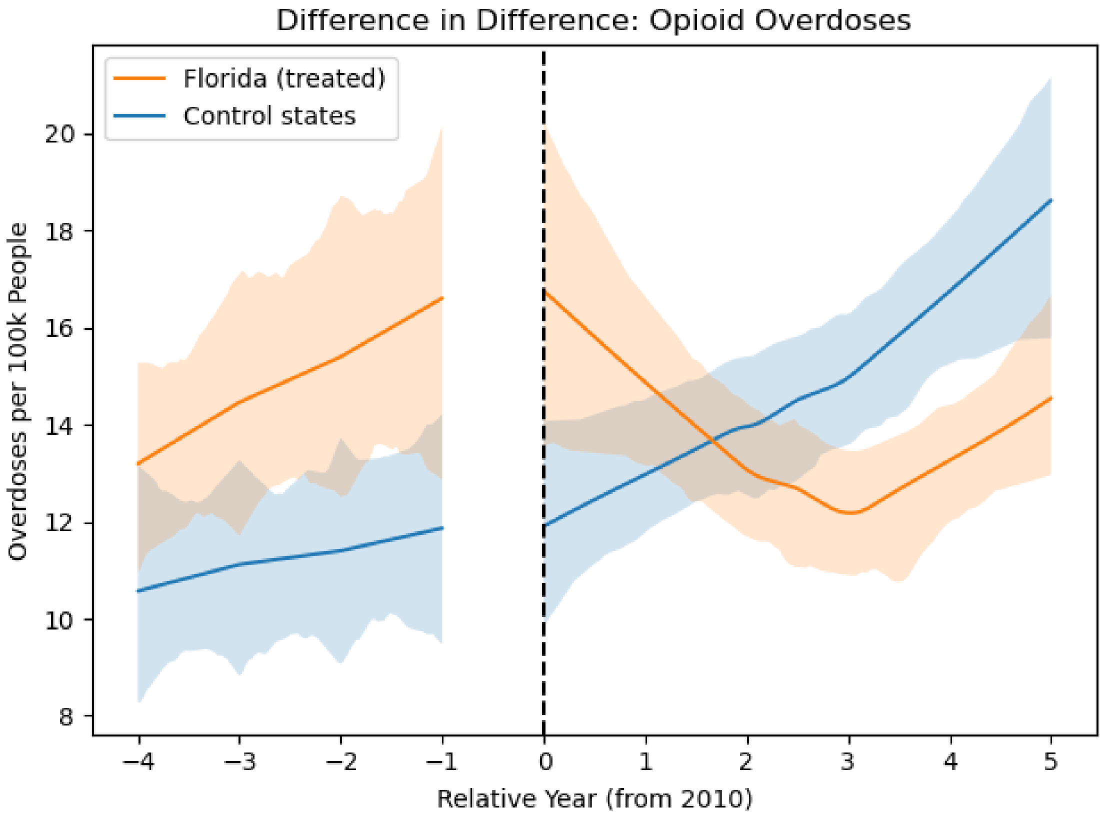

Figure 2. Difference in Difference: Opioid Overdoses

Figure 2 shows the change in opioid overdose deaths in Florida compared to the control states following the 2010 prescribing regulations.

In the first few years after the policy (relative years 0 to 2), Florida's overdose deaths decline while deaths in the control states continue to rise. This suggests that the restrictions may have produced short-term reductions in overdose mortality by reducing access to high-risk prescription opioids.

Beginning around years 3 to 5 post-policy, overdose deaths in Florida start increasing again. However, deaths also continue to rise in the control states during this time, and Florida remains below the control states for most of the post-policy period. This indicates that the initial decline did not fully disappear. Instead, Florida's outcomes improved relative to what we see in the comparison states, even though rising national trends, likely linked to the spread of illicit fentanyl, began increasing overdose deaths everywhere.

Overall, the results show that Florida's regulations were associated with a meaningful reduction in overdose deaths compared to similar states, although the effect weakened over time as other forces in the opioid crisis began to dominate.

0.7 Conclusion

Florida's 2010 opioid-prescribing regulations offer a useful case study for understanding how prescription policies can reduce opioid use and abuse. Our analysis shows that these regulations produced a large and immediate drop in prescription opioid shipments, which suggests that prescribing rules can be an effective tool for limiting the supply of opioids.

The effects on overdose deaths are more mixed. Florida's overdose mortality fell in the first few years after the policy, while deaths in similar states continued to rise. Although overdose deaths in Florida later increased again, this may be attributed to the spread of illicit opioids like fentanyl. For most of the post-policy period, Florida's overdose rate remained below that of the control states, indicating that the regulations did have a meaningful effect, even if the benefit weakened over time with the rise of other illegal opioids.

Overall, Florida's experience shows that prescribing regulations can reduce opioid availability and help decrease overdose deaths, but they are not enough on their own to produce lasting reductions in mortality. Policymakers should view these regulations as one part of a larger strategy. The US DEA, US HHS, and state-level agencies should combine prescription regulations with other policies to be more effective at reducing opioid-related deaths over the long term.

0.8 Appendix

The figures below present pre–post comparisons for Florida, showing how prescription opioid shipments and overdose mortality changed before and after the 2010 policy implementation. These plots are included for completeness, as pre–post comparisons form the foundation for the difference-in-difference approach used in the main analysis.

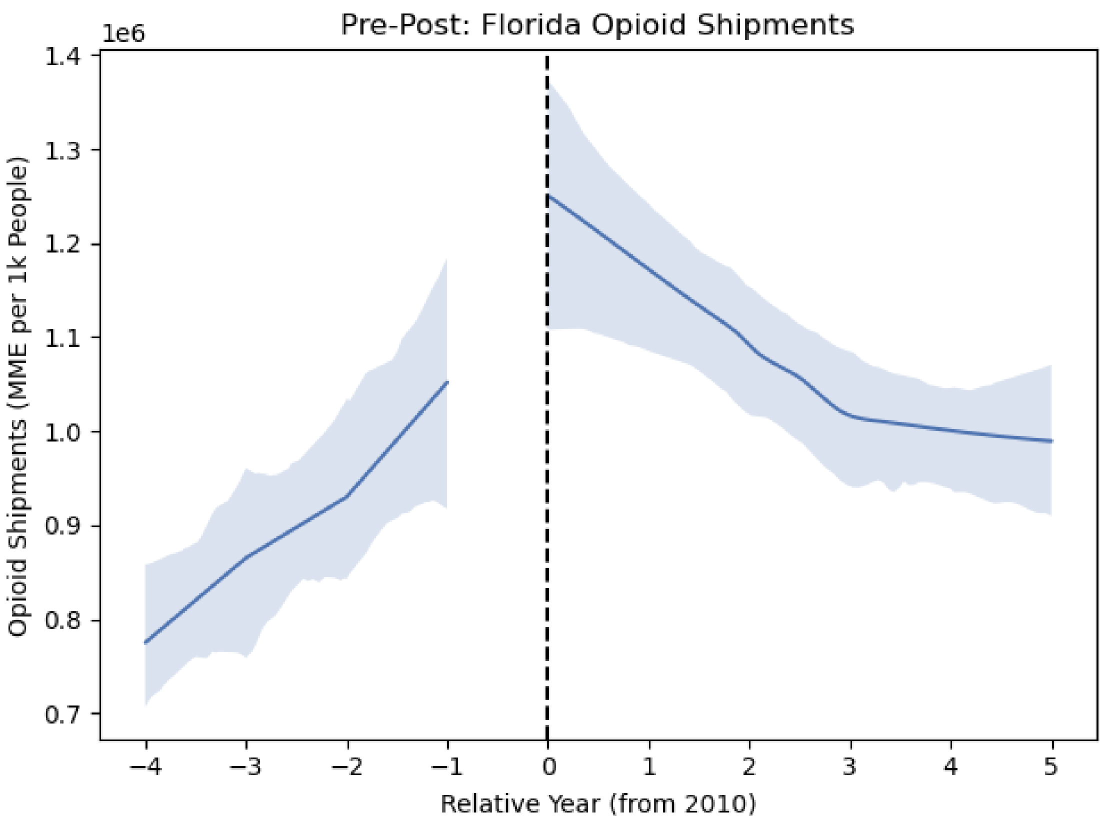

Figure 3. Pre-Post: Opioid Shipments

This figure shows prescription opioid shipments (measured in MME per 1,000 people) in Florida from 2006 to 2015. Shipments rise steadily from 2006 to 2009, then decline sharply after the 2010 regulations. The pre–post pattern suggests a large reduction in the supply of prescription opioids following the policy change.

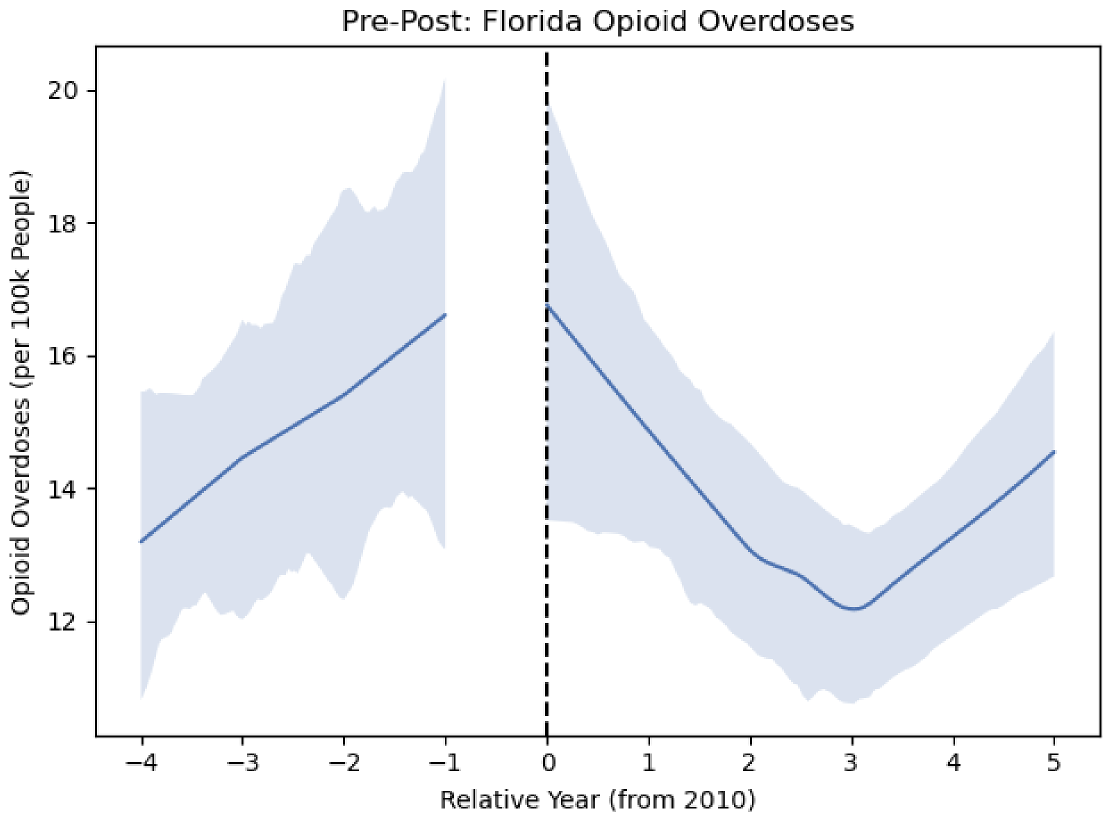

Figure 4. Pre-Post: Opioid Overdoses

This figure shows Florida's overdose deaths per 100,000 people from 2006 to 2015. Overdose mortality rises steadily prior to 2010, decreases briefly in the first three years after the policy, and then begins rising again. This pattern is the need for a comparison group, as statewide or national factors unrelated to Florida's policy could be influencing these trends.

Choosing Control States: Before comparing overdose or prescription patterns, we first explored the states that had similar population size to Florida prior to 2010 since we assume opioid deaths and prescription shipments tend to scale with population size. For each state, we calculated the average population for each year between 2006 and 2009 and compared it to Florida's average population during the same period. We used different thresholds for selecting states, such as those with populations within $\pm 10\%$, $\pm 15\%$, and $\pm 30\%$ of Florida's average population.

Next, we compared how overdose rates and prescription shipment trends changed over time from 2006 to 2009 (pre-policy). It's reasonable for us to conclude that states whose year-to-year differences move in similar ways to Florida's may act as good control states, since they exhibit similar trends. We calculated how overdose rates and prescription shipments changed from year to year for the years 2006 to 2009 for Florida and all other states in our dataframe. States that had similar trends (i.e. slopes) to Florida for both overdose rates and prescription shipments were considered the strongest controls.

There weren't many states that met all requirements, so in our pursuit for identifying control states, we prioritized those with trends most similar to Florida over similar average population size. The states with similar upward prescription shipment trends were Ohio, Alabama, Georgia, and Kentucky. The latter three states are also geographically similar to Florida. The states with similar increasing overdose death trends were Colorado, Michigan, and Pennsylvania. While these states are less geographically and culturally similar to Florida, our analysis finds that they display the closest parallel to Florida's trends.